

Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal
Lecture: CPR 50

**Title: Acid Base Balance: Buffers and
Respiratory System**

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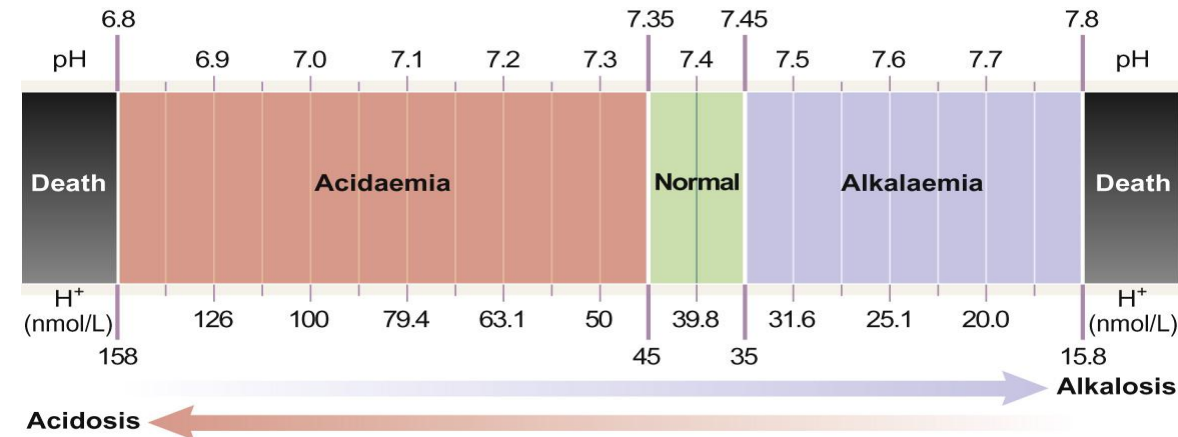
Lecture: Acid Base Balance: Buffers and Respiratory System

SOM.MK.I.BPM1.3.CPR.1.INTG. 0001	Define an acid and a base and distinguish between strong and weak acids and bases. Define pH, pKa and buffers.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0003	Analyze the titration curve of a buffer (identify the pK and regions of maximum buffering capacity of a buffer) using the Henderson-Hasselbalch equation.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0004	Distinguish the sources, amounts formed and routes of loss of: volatile acid (CO ₂ derived) and nonvolatile acids
SOM.MK.I.BPM1.3.CPR.1.INTG. 0005	Analyze the role of HCO ₃ ⁻ as a buffer (its formation and role in buffering) using a graph.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0006	Identify the normal range of pH values as well as the upper and lower limits compatible with life. Emphasize the importance of the 20:1 ratio for maintenance of plasma pH.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0007	Describe the role of hemoglobin as a buffer in CO ₂ transport
SOM.MK.I.BPM1.3.CPR.1.INTG. 0008	Summarize the role of lungs in maintaining the arterial PCO ₂ .

Recommended Reading

- Medical Physiology: Principles for Clinical Medicine, R. Rhoades & D. Bell; 6th ed.; Chapter 24, pp. 533-555; [Acid–Base Homeostasis | Medical Physiology: Principles for Clinical Medicine, 6e | Premium Basic Sciences | Health Library \(oclc.org\)](#)
- Costanzo Physiology; <https://www.clinicalkey.com/student/content/book/3-s2.0-B978032379333900016X>
- Ref: Mark's Essentials of Biochemistry: Chapter 4; [Water, Acids, Bases, and Buffers | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Premium Basic Sciences | Health Library \(oclc.org\)](#)
- Reference: Vander's Renal Physiology; Chapter 9: Acid base balance; [Regulation of Acid-Base Balance | Vander's Renal Physiology, 10th Edition | AccessMedicine | McGraw Hill Medical \(oclc.org\)](#)
- Additional problems: <http://www.scribd.com/doc/6442924/ABG-Interpretation>
Slide 17 onwards
- http://www.mhhe.com/biosci/ap/ap_casestudies/cases/ap_case14.html
- Practice questions folder for the week; Small group case

Acids and bases



- Acids can donate hydrogen ion (H^+) / accepts electrons
 - Acidosis can be due to addition of acid
 - Alkalosis can be due to loss of acid
- Bases accept hydrogen ions/ donates electrons
 - Acidosis can be due to loss of base
 - Alkalosis can be due to gain of base
- Strong acids dissociate completely; Weak acids dissociate to a limited extent (suitable as buffers)
- pH is **negative log of $[H^+]$** ; At a plasma pH = 7.4, the $[H^+] = 35-45$ nmol/L
- Note: $[H^+]$ and pH inversely related, and pH is logarithmic scale

Acid base balance

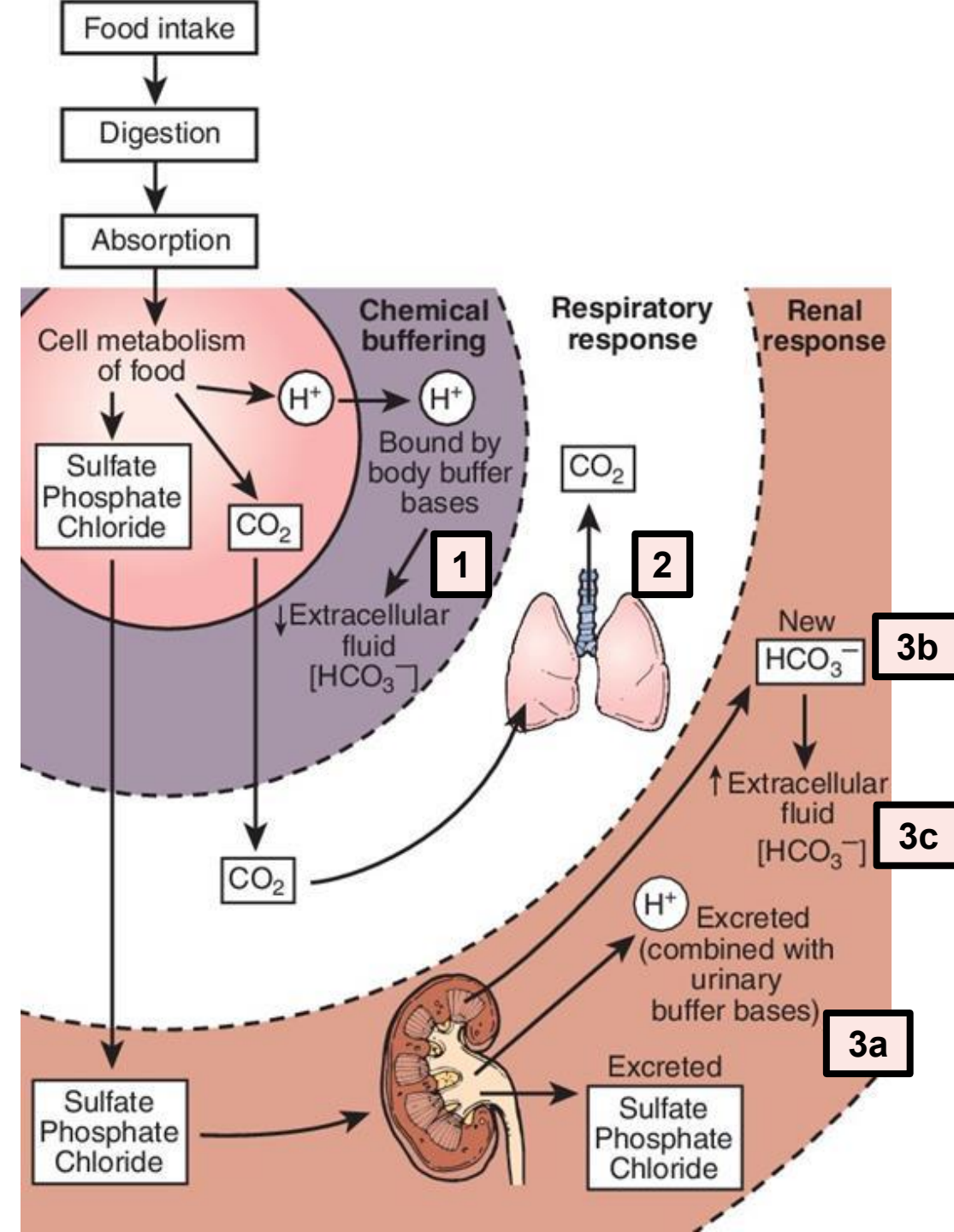
- Plasma pH = 7.4 (± 0.05); $[H^+] = 35-45 \text{ nmol/L}$
- Note logarithmic relation and inverse relation between pH and $[H^+]$
- Intracellular enzymes and protein functions are pH dependent
- Metabolism produces several acids (about 20,000 mmol/day)
- Three systems to regulate $[H^+]$:
 - Buffers (First line) – Immediate defense
 - Respiratory system (Second line) – Regulates PCO_2
 - Renal system (Third line) – Regulates HCO_3^- levels

Metabolic production of acids

- Volatile acid (Carbonic acid or carbon dioxide or CO_2)
 - Major metabolic acid (15,000-20,000 mEq/day)
 - Handled by lungs
- **Nonvolatile/ fixed acids** (50-100 mEq/day)
 - Handled by the kidneys
 - Inorganic acids
 - Sulfuric acid (H_2SO_4) formed by sulfur containing amino acid metabolism
 - Phosphoric acid (H_3PO_4) formed by metabolism of phospholipids...
 - Organic acids
 - Ketone bodies (acetoacetic acid, 3-hydroxybutyric acid)
 - Lactic acid (glycolysis under anaerobic conditions)
 - **Overproduction/accumulation causes metabolic acidosis**
- How is the plasma pH ($[\text{H}^+]=35\text{-}45\text{ nmol/L}$) maintained, even if acid is added????

Acid base Homeostasis

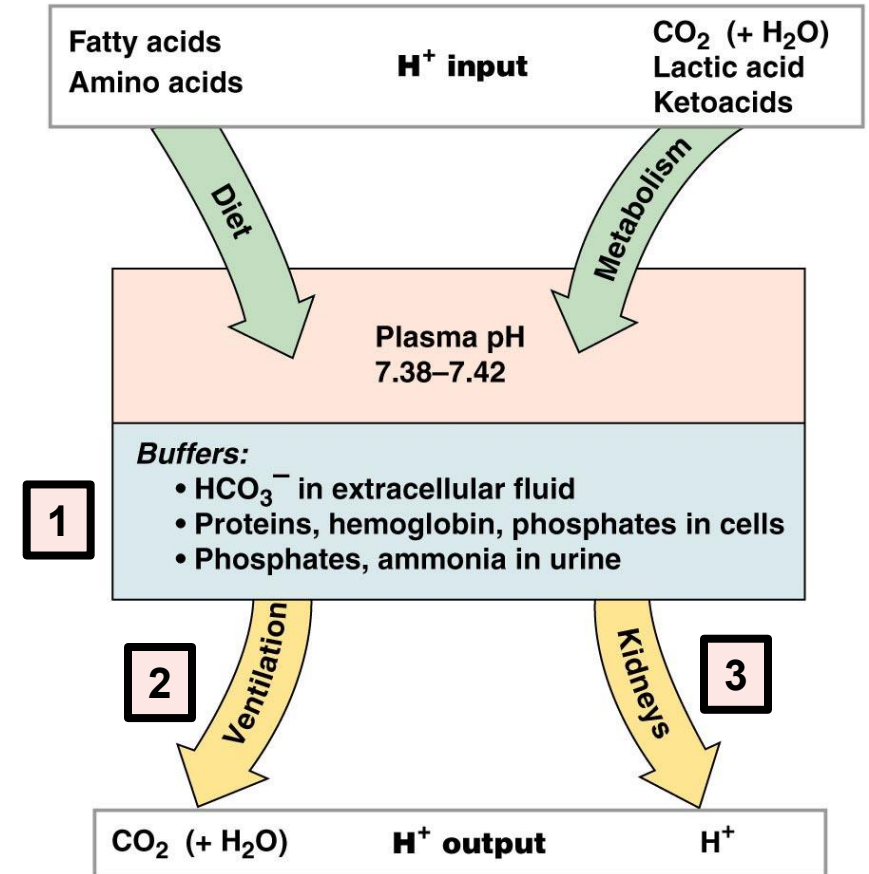
- Buffers, respiratory system, and kidneys
 - On a mixed diet, fixed acids produced (sulfuric, hydrochloric, and phosphoric) because of protein metabolism.
1. Acids buffered by **ECF HCO_3^-**
 2. **Respiratory system** disposes off CO_2
 3. **Kidneys**
 - a. **Eliminate H^+** (combined with urinary buffers) and anions (sulfate/phosphate) in urine
 - b. Add **new HCO_3^-** to ECF to replace HCO_3^- consumed in buffering strong acids.
 - c. **Reabsorb filtered HCO_3^-**



Defenses Against Changes in $[H^+]$

Three primary systems to regulate H^+ :

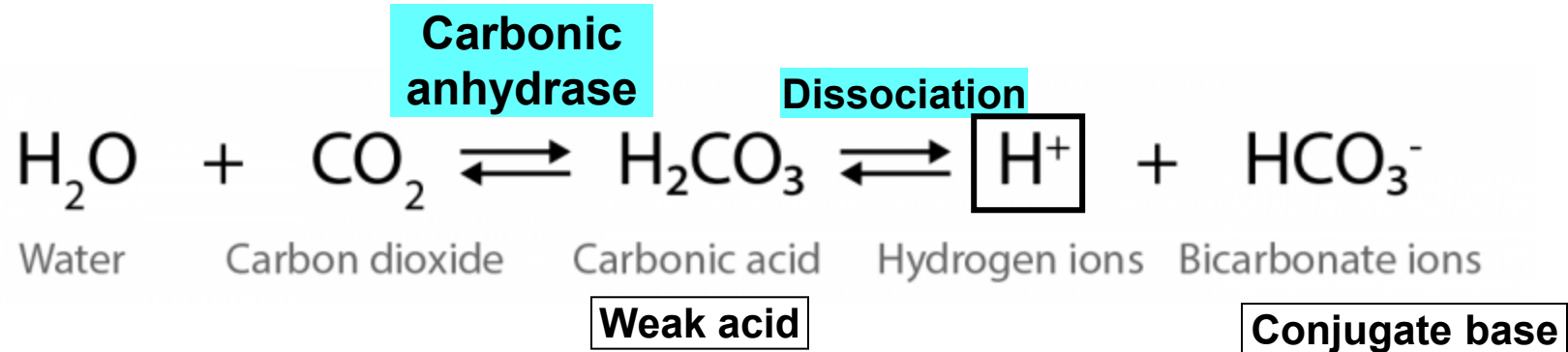
1. Buffering systems (ECF and ICF):
IMMEDIATELY combine with acid/base to prevent large changes in $[H^+]$ and pH
2. Respiratory Response:
Within minutes to eliminate CO_2 (H_2CO_3)
3. Renal Response:
Slowest - hours/days to eliminate excess acid/base
Regulates HCO_3^-
MOST POWERFUL REGULATORY SYSTEM



Body buffers: First line of defense

- Plasma pH: 7.35-7.45 and intracellular pH is about 7.1
- Buffers **minimize change in pH (do NOT prevent it)** on addition of small quantity of acid or base
- Major body buffer systems
 - **Bicarbonate-carbonic acid buffer (ECF buffer)**
 - Hemoglobin (red blood cells) – due to **histidine** residues
 - Phosphate buffer (ICF buffer and **renal tubular buffer**)
 - Proteins (ICF and plasma) - due to histidine residues
 - Ammonia buffer (**renal tubule**)

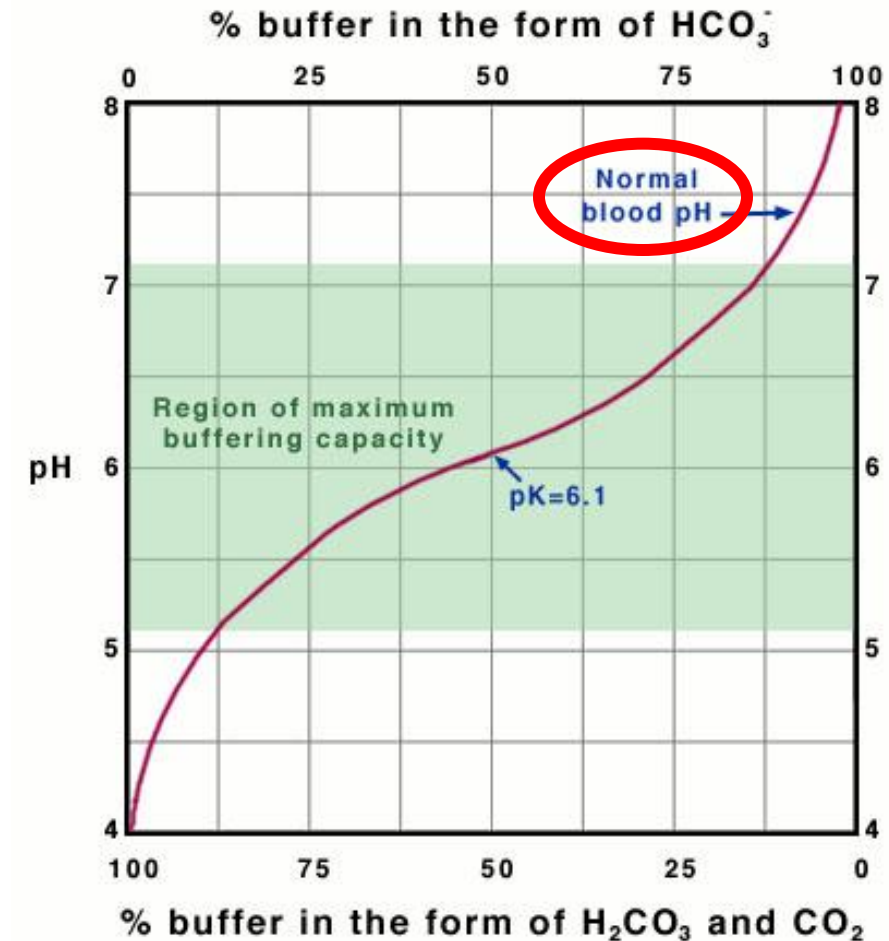
Bicarbonate buffer: Major ECF buffer



- Carbonic anhydrase (reversible)
 - CO_2 + water forms carbonic acid (H_2CO_3)
 - Different enzyme isoforms in RBC, renal tubule, parietal cells in stomach
- Weak acid ($\text{HA}/ \text{H}_2\text{CO}_3/ \text{CO}_2$) and its conjugate base ($\text{A}^-/ \text{HCO}_3^-$)
- Conjugate pair = acid and base that differ by the presence of a proton
- Addition of H^+ , forms $\text{H}_2\text{CO}_3 \rightarrow \rightarrow \text{CO}_2$ lost via lungs
 - Small amount of bicarbonate is lost by buffering
- When $\text{pH} = \text{pK}_a$ (6.1), **[weak acid] = [conjugate base]**

Bicarbonate buffer: Major ECF buffer

- **Effectiveness of bicarbonate buffer:**
 - pK_a is 6.1 (plasma pH = 7.4):
 - **[weak acid] = [conjugate base]**
 - Effective range: 5.1-7.1
 - **Slightly outside the buffering range**
 - **High concentration** in ECF
- At pH = pK_a (6.1), **[weak acid] = [conjugate base]**
- At pH = 5.1, [base]: [acid] = **1:10**
 - When pH = 4.1, [base]: [acid] = **1:100**
- At pH = 7.1, ratio of [base]: [acid] = **10:1**



Bicarbonate buffer

- In clinical setting, apply Henderson-Hasselbalch equation for bicarbonate buffer
- pKa of bicarbonate buffer is **6.1**
- Serum $[\text{HCO}_3^-] = 22\text{-}26 \text{ mEq/L}$ (24 mEq/L)
- Arterial $\text{PaCO}_2 = 38\text{-}42 \text{ mmHg}$ (40 mmHg)
- Plasma pH of 7.4, ratio of [Base]/[Acid] is **20:1**

$$\text{pH} = \text{pK}_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

Bicarbonate buffer

$$\text{pH} = \text{pK}_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

$$\text{pH} = \text{pK}_a + \log \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]}$$

But $[\text{H}_2\text{CO}_3]$ - very low and difficult to measure;

$$[\text{H}_2\text{CO}_3] = 0.03 \times \text{PCO}_2 \text{ (Dissolved CO}_2\text{)}$$

$0.03 = \alpha$ = Solubility coefficient (constant)

$$\text{pH} = 6.1 + \log \frac{[\text{HCO}_3^-]}{0.03 \times \text{PCO}_2}$$

$$\text{pH} = 6.1 + \log \frac{24}{0.03 \times 40}$$

Plasma pH of 7.4, ratio of **[Base]/[Acid]** (24:1.2) = **20:1**

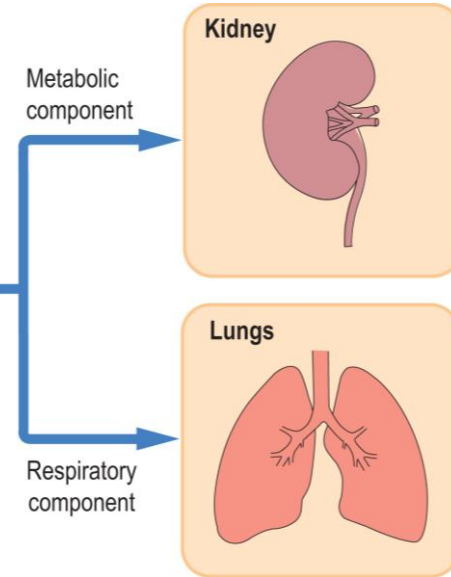
Bicarbonate buffer

$$\text{pH} = \text{pK}_a + \log \frac{[\text{HCO}_3^-]}{0.03 \times \text{PaCO}_2}$$

Bicarbonate levels
Renal system

PaCO₂ levels
Respiratory system

$$\text{pH} = \frac{[\text{HCO}_3^-]}{\text{PCO}_2}$$



- **Most effective ECF buffer:**
 - pK_a is 6.1 (plasma pH = 7.4): Slightly outside effective buffering range
 - **High concentration** in ECF
 - **Bicarbonate levels** and **PCO₂** **independently regulated** by two different systems (**open system**)
- Assess acid-base status: Measure arterial pH, PaCO₂ and HCO₃⁻ (Davenport diagram)

Central Message of Henderson-Hasselbalch equation

$$\text{pH} = \text{pKa} + \log(\text{HCO}_3^- / 0.03 \times \text{PaCO}_2)$$

0.03 = α = Solubility coefficient (constant)

Plasma pH = Function of $\text{HCO}_3^- : \alpha \text{PaCO}_2$ ratio (20:1)

$\uparrow (\text{HCO}_3^- : \alpha \text{PCO}_2) = \text{Alkalosis}$

Could be due to $\text{HCO}_3^- \uparrow$ (Metabolic) or $\text{PCO}_2 \downarrow$ (Respiratory)

$\downarrow (\text{HCO}_3^- : \alpha \text{PCO}_2) = \text{Acidosis}$

Could be due to $\text{HCO}_3^- \downarrow$ (Metabolic) or $\text{PCO}_2 \uparrow$ (Respiratory)

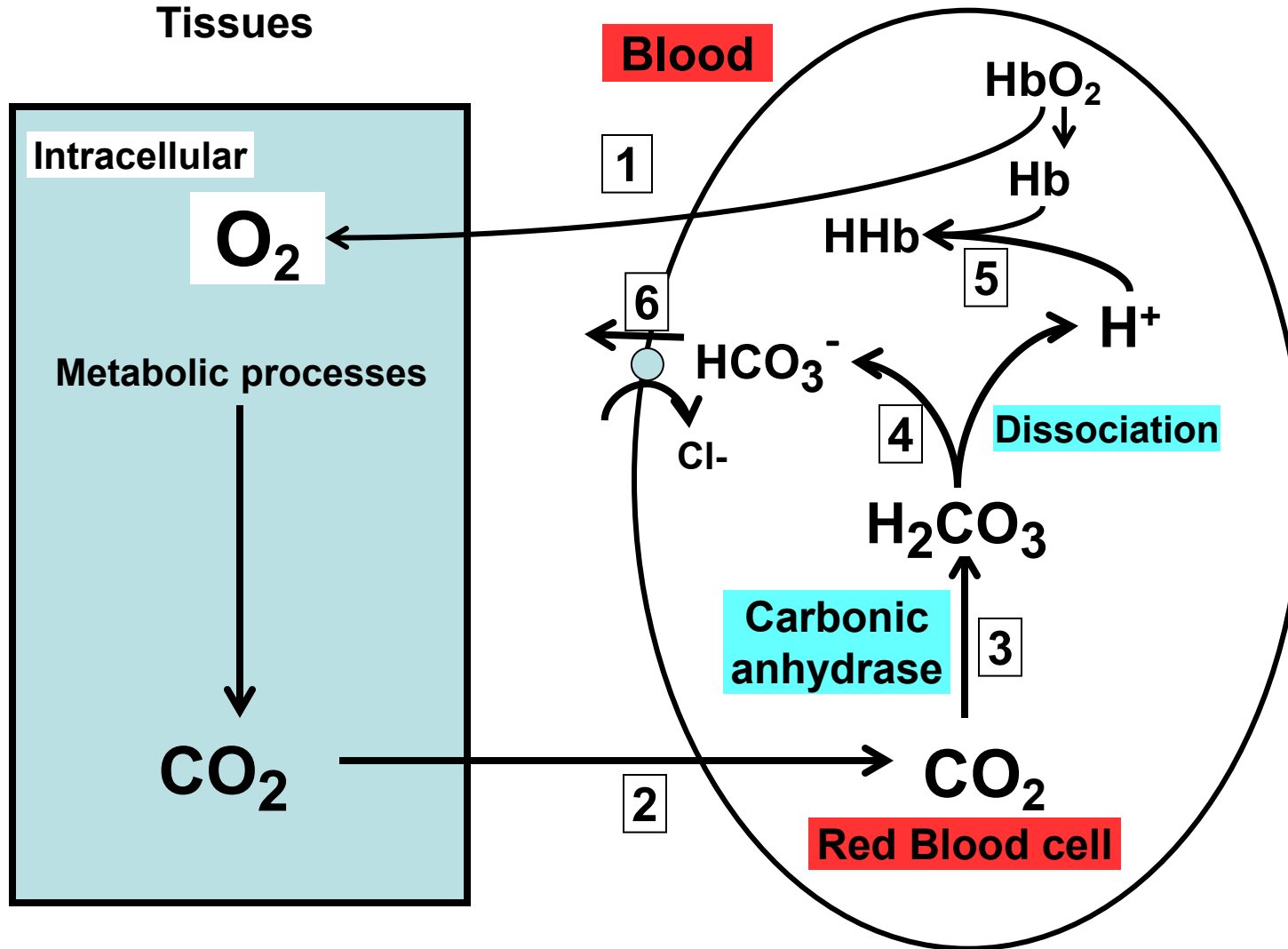
Effect of Changes in PCO_2 , HCO_3^- and ratio of $\text{HCO}_3^-/\alpha\text{PCO}_2$

			Normal	
PCO_2 held constant	PCO_2	40	40	40
	HCO_3^-	48	24	12
	$\text{HCO}_3^-/\alpha\text{PCO}_2$	40	20	10
	pH	7.7	7.4	7.1
	PCO_2	20	40	80
HCO_3^- held constant	HCO_3^-	24	24	24
	$\text{HCO}_3^-/\alpha\text{PCO}_2$	40	20	10
	pH	7.7	7.4	7.1
	PCO_2	80	40	20
	HCO_3^-	48	24	12
$\text{HCO}_3^-/\alpha\text{PCO}_2$ constant	$\text{HCO}_3^-/\alpha\text{PCO}_2$	20	20	20
	pH	7.4	7.4	7.4

Handling of volatile acid CO_2

- CO_2 production about 15,000-20,000 mEq/day
 - Produced by carbohydrate, fat and amino acid metabolism
- CO_2 lost via the lungs (Volatile acid)
- Respiratory system regulates PaCO_2 (**acid component**) of bicarbonate buffer system
- **Hemoglobin** (non-bicarbonate buffers) accepts H^+ formed during CO_2 transport from tissues (venous blood pH slightly lower than arterial)
- About 70-75% of CO_2 transported as HCO_3^-
 - **HCO_3^- in venous blood higher than in arterial blood** (explain)
 - Rest is as carbamino hemoglobin and dissolved CO_2
- Carbonic anhydrase is rich in RBC

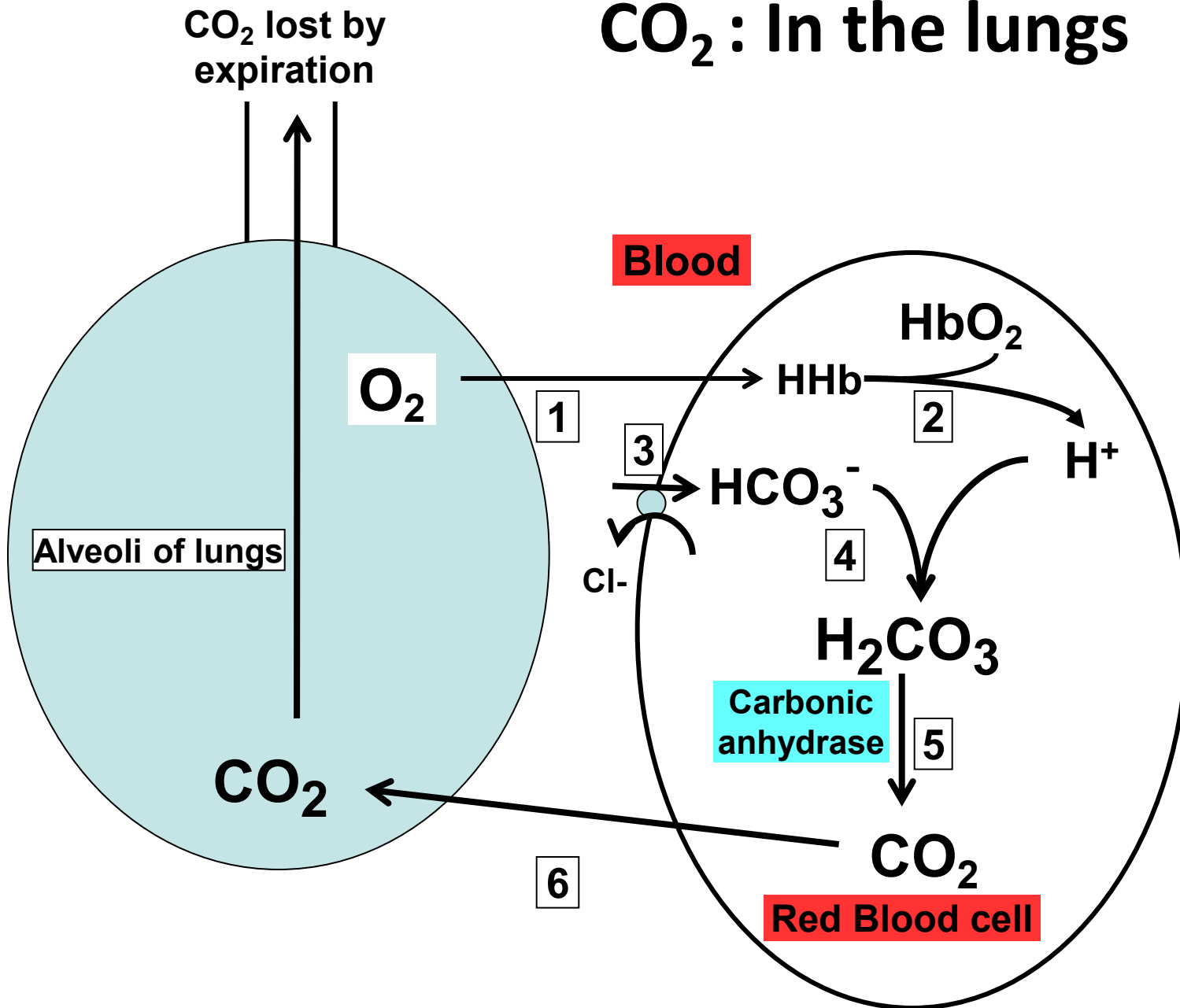
Transport of CO₂ from tissues to lungs



1. Oxygen dissociates from Hemoglobin → DeoxyHb (T state)
2. CO₂ diffuses from tissues into blood (RBC)
3. Formation of carbonic acid in presence of carbonic anhydrase
4. Carbonic acid (weak acid) dissociates
5. Hydrogen ions formed buffered by hemoglobin (**histidine**): T state
6. Chloride shift
7. HCO₃⁻ in venous blood higher than in arterial blood

**Hemoglobin buffer
accepts H⁺ formed by CO₂**

CO₂ : In the lungs

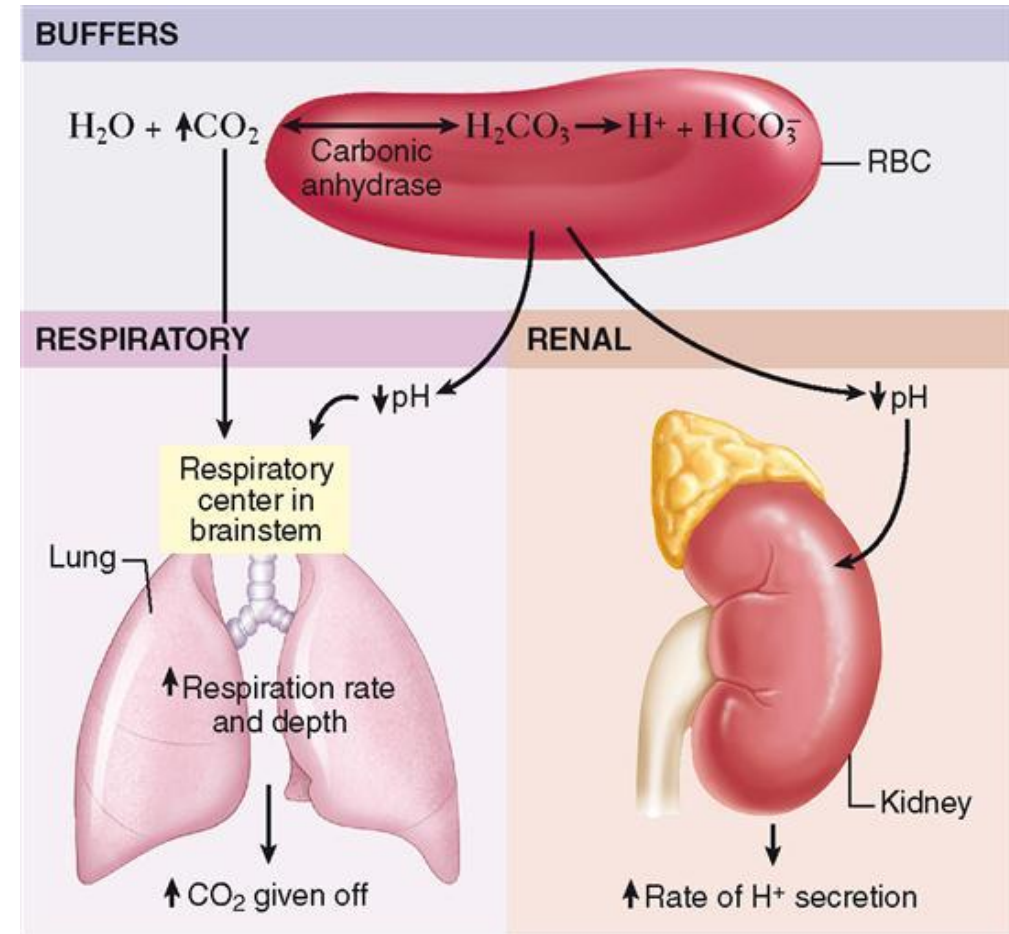
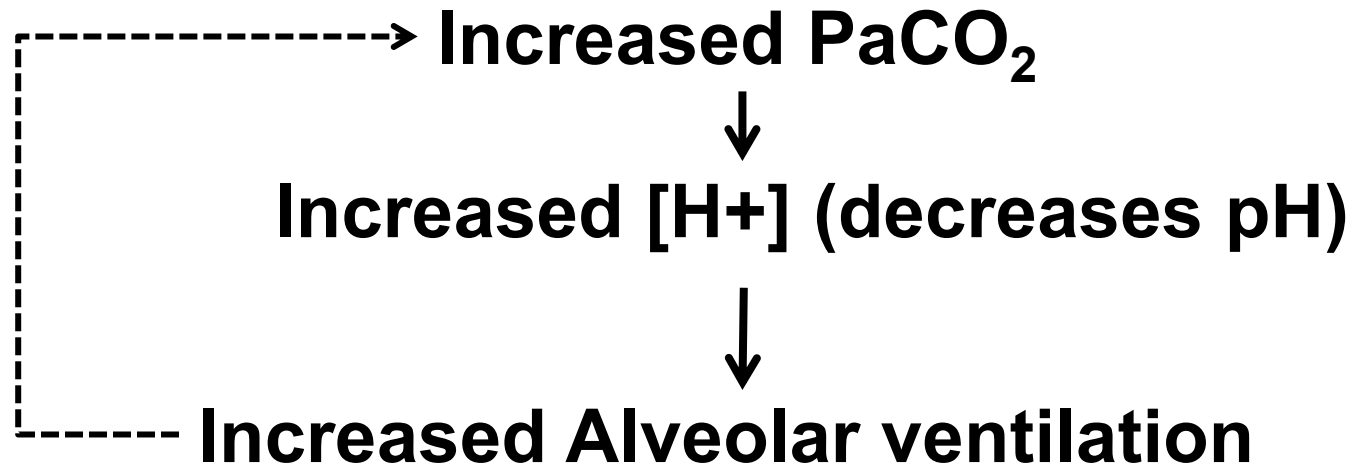


1. Oxygen diffuses from alveoli into RBC and binds Hb → OxyHb (R state)
2. H⁺ released from histidine residues of Hb
3. HCO₃⁻ enters RBC in exchange for Cl⁻
3. H⁺ associate with HCO₃⁻ to form carbonic acid (H₂CO₃)
4. CO₂ formed from carbonic acid by **carbonic anhydrase** (reversible)
5. CO₂ diffuses into alveoli and is lost by expiration

Respiration Relating to PCO_2 and pH

Chemoreceptors: Increased $\text{PaCO}_2 \rightarrow$ Increases $[\text{H}^+]$ which stimulates respiratory center increasing respiration $\rightarrow$ Hyperventilation

Control centers respond by increasing/decreasing ventilation to keep PaCO_2 near 40 mmHg (-ve feedback controller)



Role of respiratory center in pH regulation

- Peripheral chemoreceptors respond **quickly** to changes in pH
- When blood pH falls (metabolic acidosis), stimulates respiratory center → **Increased rate and depth of respiration (hyperventilation)**. Increased CO_2 washout, **lowers PaCO_2** (**Compensatory response**)
- When blood pH increases (metabolic alkalosis), inhibits respiratory center → **Decrease in respiratory rate (hypoventilation)**. Increased CO_2 retention, **increases PaCO_2** (**Compensatory response**)
- Respiratory system **regulates PaCO_2** (acid) component of bicarbonate buffer

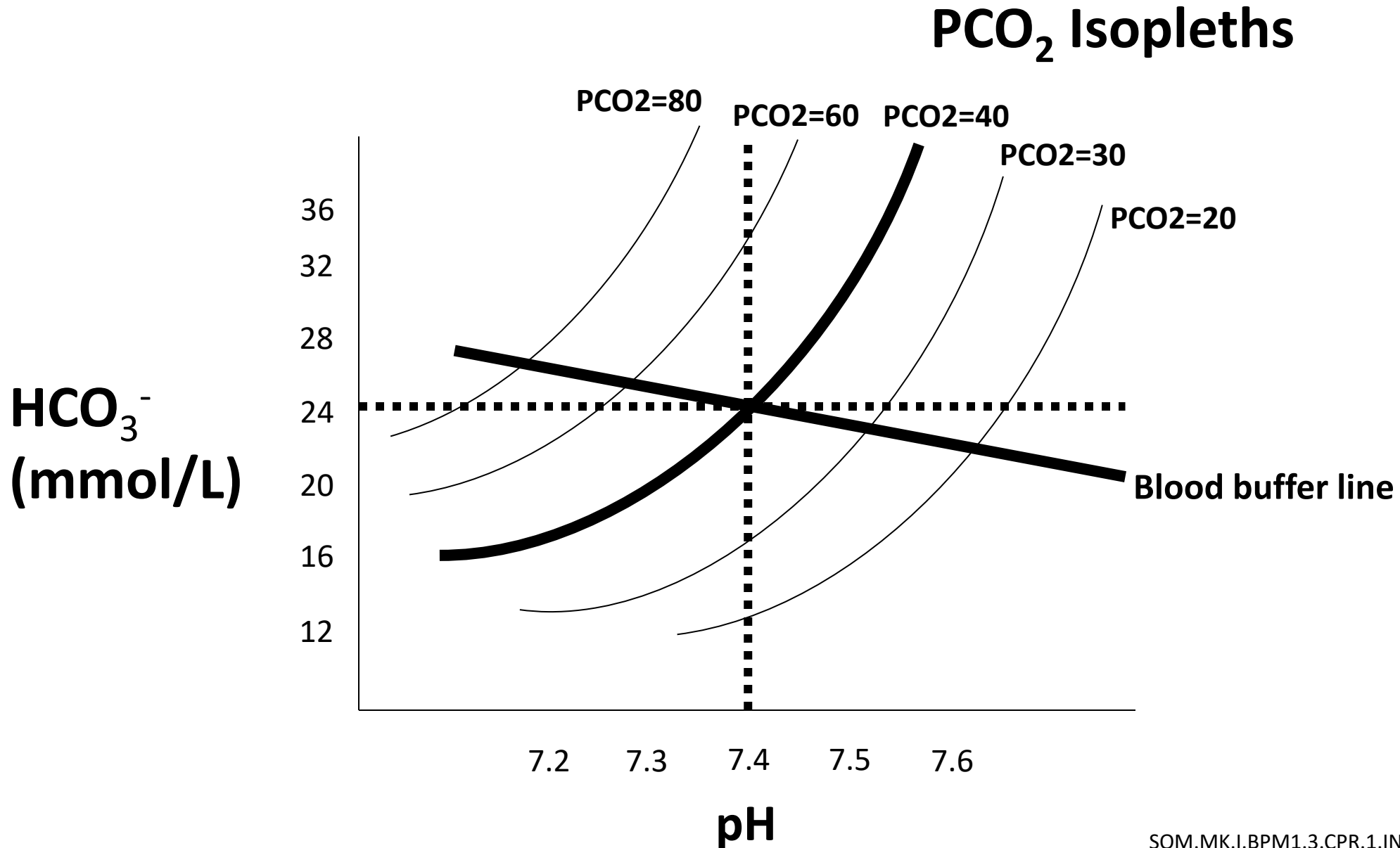
Respiratory acid base disorders

- When CO_2 production = CO_2 loss via expiration, arterial PaCO_2 maintained constant
- Disorders which **decrease alveolar ventilation** result in **CO_2 accumulation (respiratory acidosis)** and **increase in PaCO_2**
 $\uparrow \text{CO}_2 + \text{H}_2\text{O} \leftrightarrow \text{H}_2\text{CO}_3 \leftrightarrow \uparrow \text{H}^+ + \text{HCO}_3^-$
- Disorders which increase alveolar ventilation result in **washout of CO_2 (respiratory alkalosis)** and a **fall in PaCO_2**
 $\downarrow \text{CO}_2 + \text{H}_2\text{O} \leftrightarrow \text{H}_2\text{CO}_3 \leftrightarrow \downarrow \text{H}^+ + \text{HCO}_3^-$

Davenport diagram

- pH-bicarbonate- PaCO_2 diagram
- Gives an approximate idea regarding the acid base disturbance

Davenport diagram



Classification of Acid-base disorders

Based on pH

Acidosis

Alkalosis

**Respiratory
Acidosis**

pH: <7.4
PaCO₂: Increased

**Metabolic
Acidosis**

pH: <7.4
HCO₃⁻: Decreased

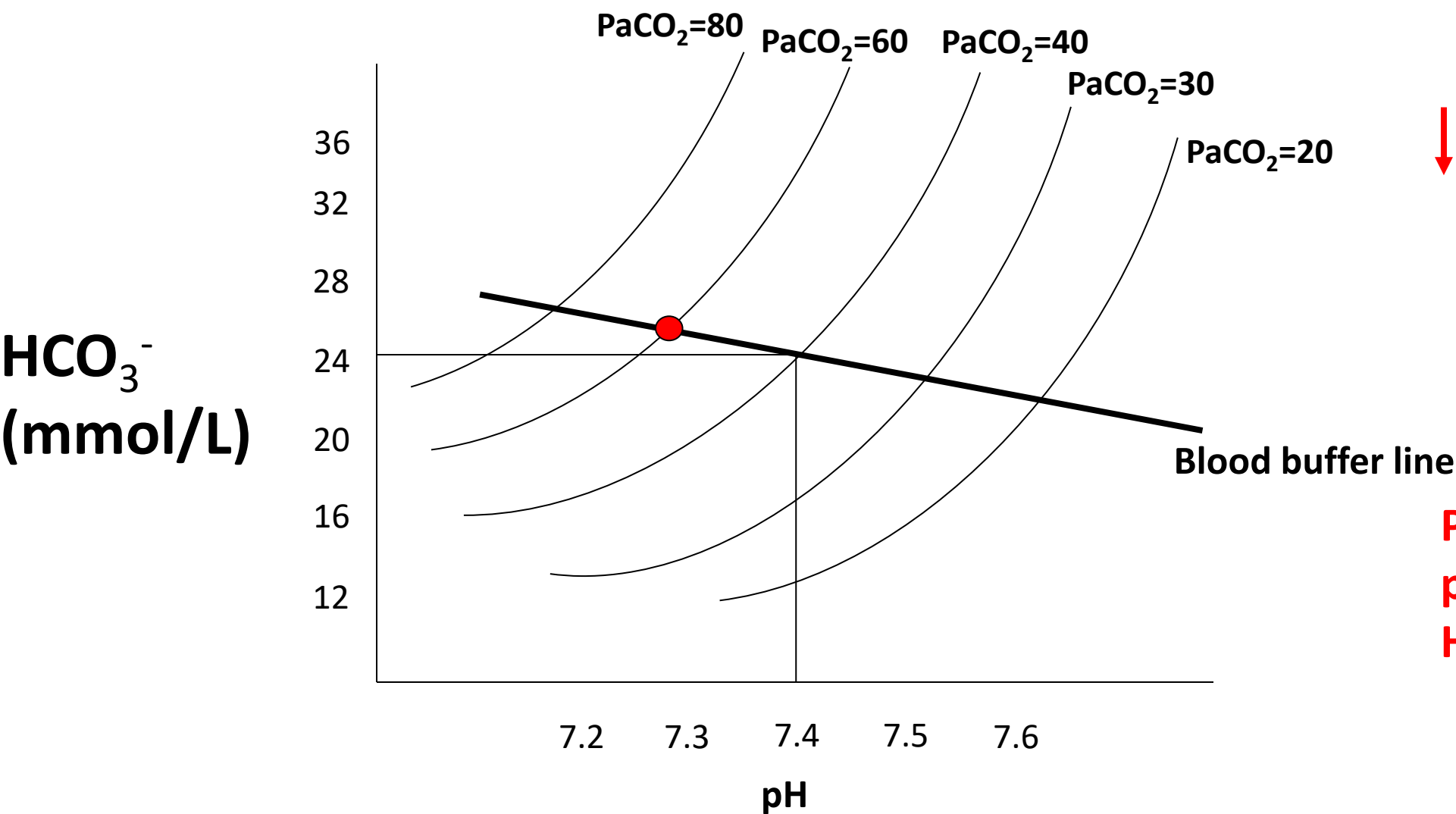
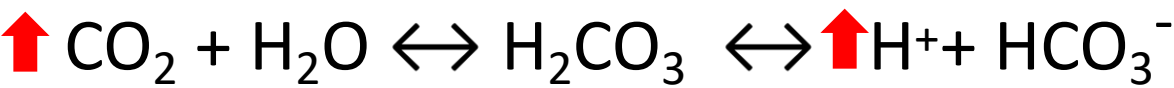
**Respiratory
Alkalosis**

pH: >7.4
PaCO₂: Decreased

**Metabolic
Alkalosis**

pH: >7.4
HCO₃⁻: Increased

Acute Respiratory acidosis ($\uparrow\uparrow \text{PaCO}_2$)

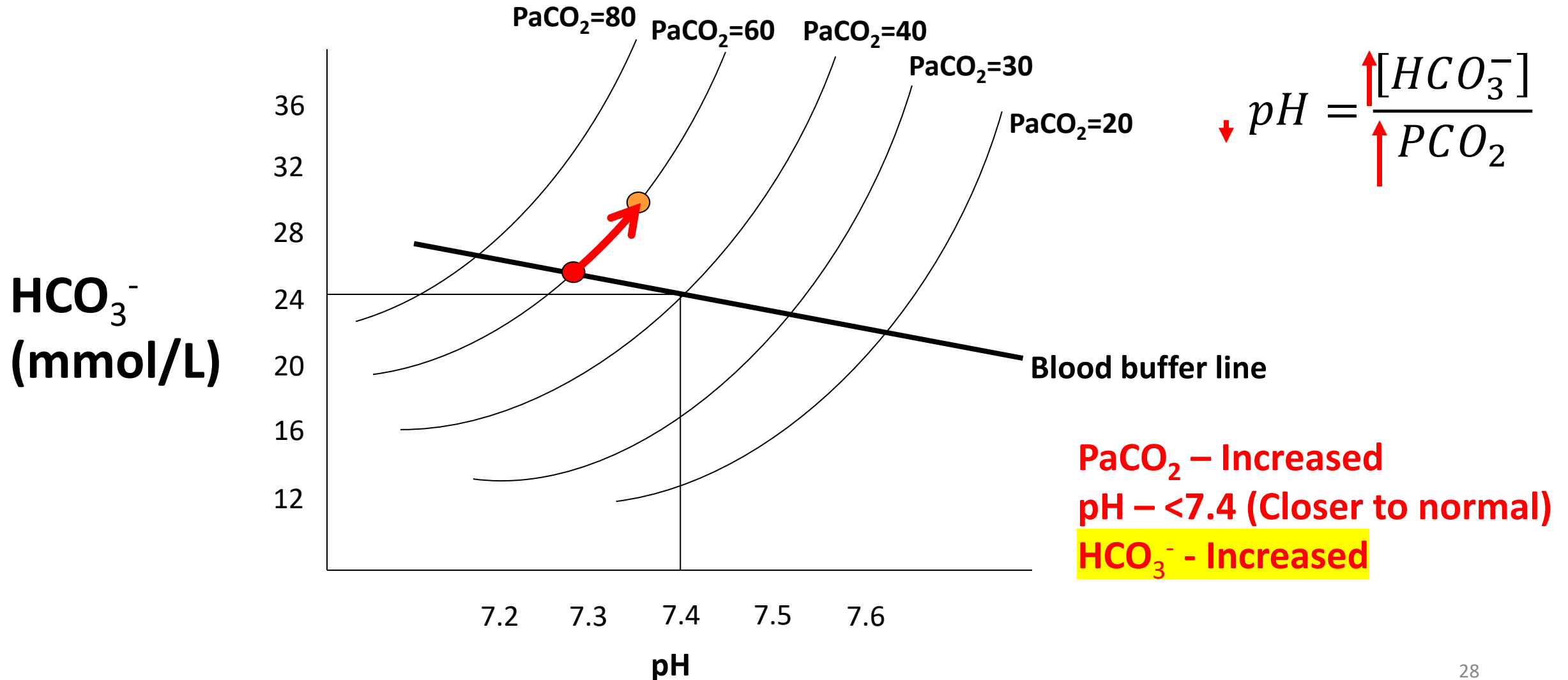


$\downarrow pH = \frac{[\text{HCO}_3^-]}{\uparrow \text{PCO}_2}$

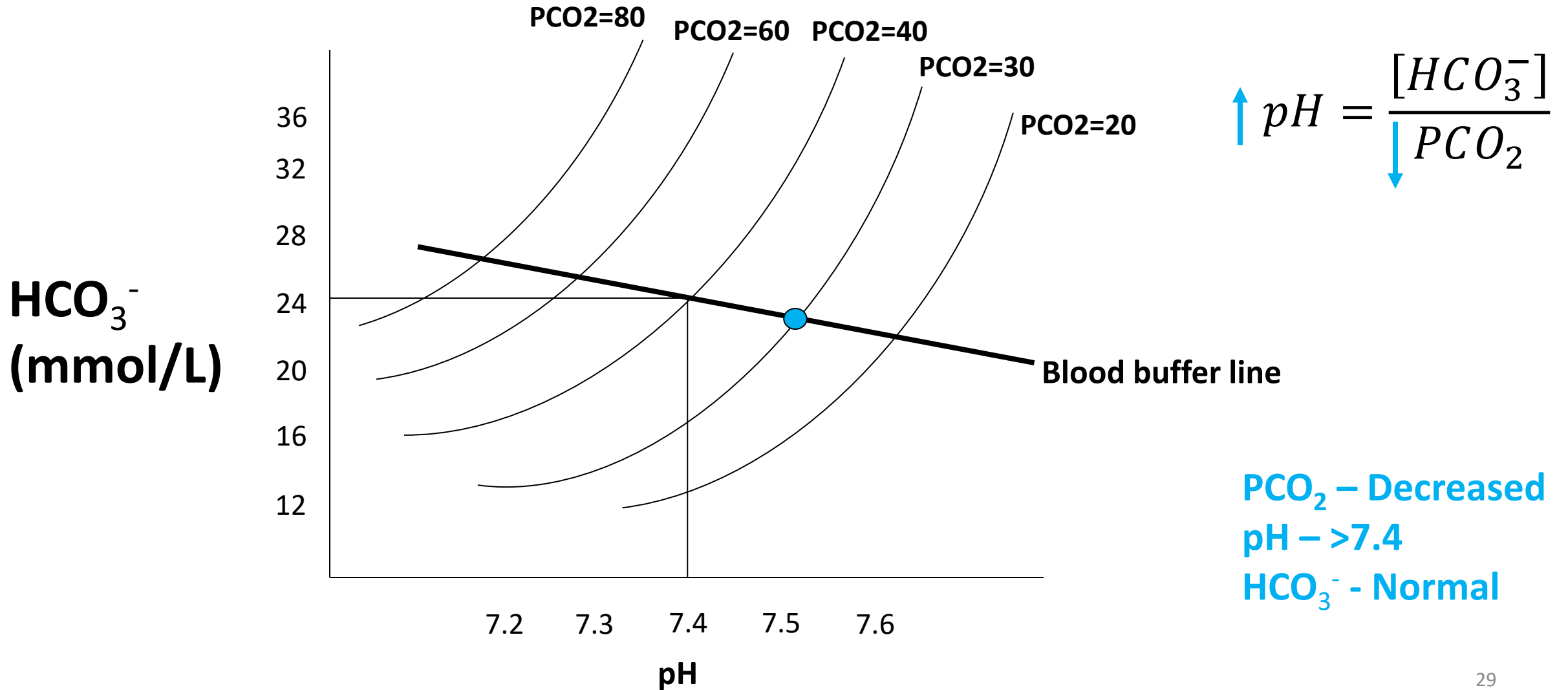
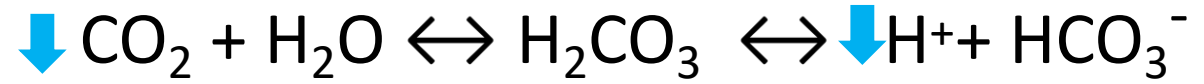
PaCO_2 – Increased
pH – <7.4
 HCO_3^- - Normal

Chronic respiratory acidosis

PCO₂ Isopleths

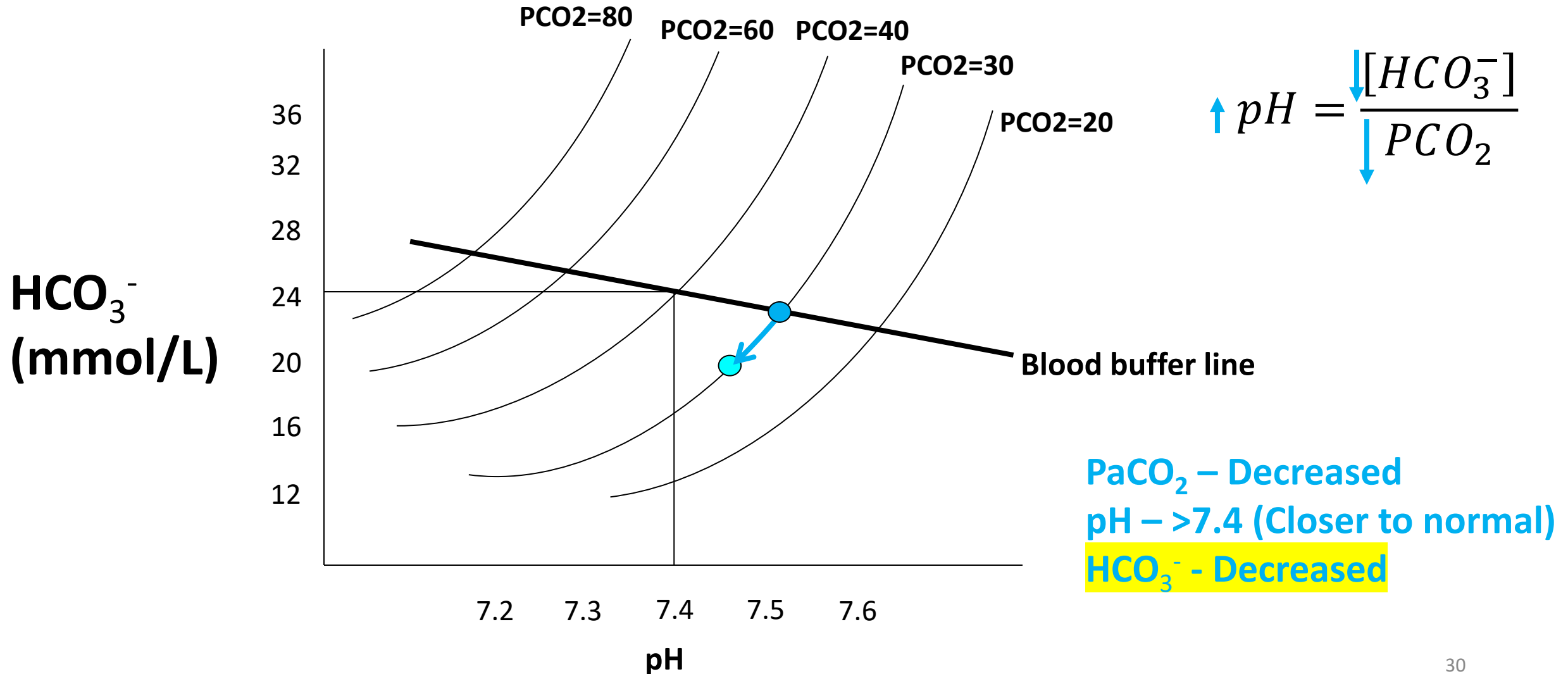


Acute respiratory alkalosis ($\downarrow\downarrow \text{PaCO}_2$)

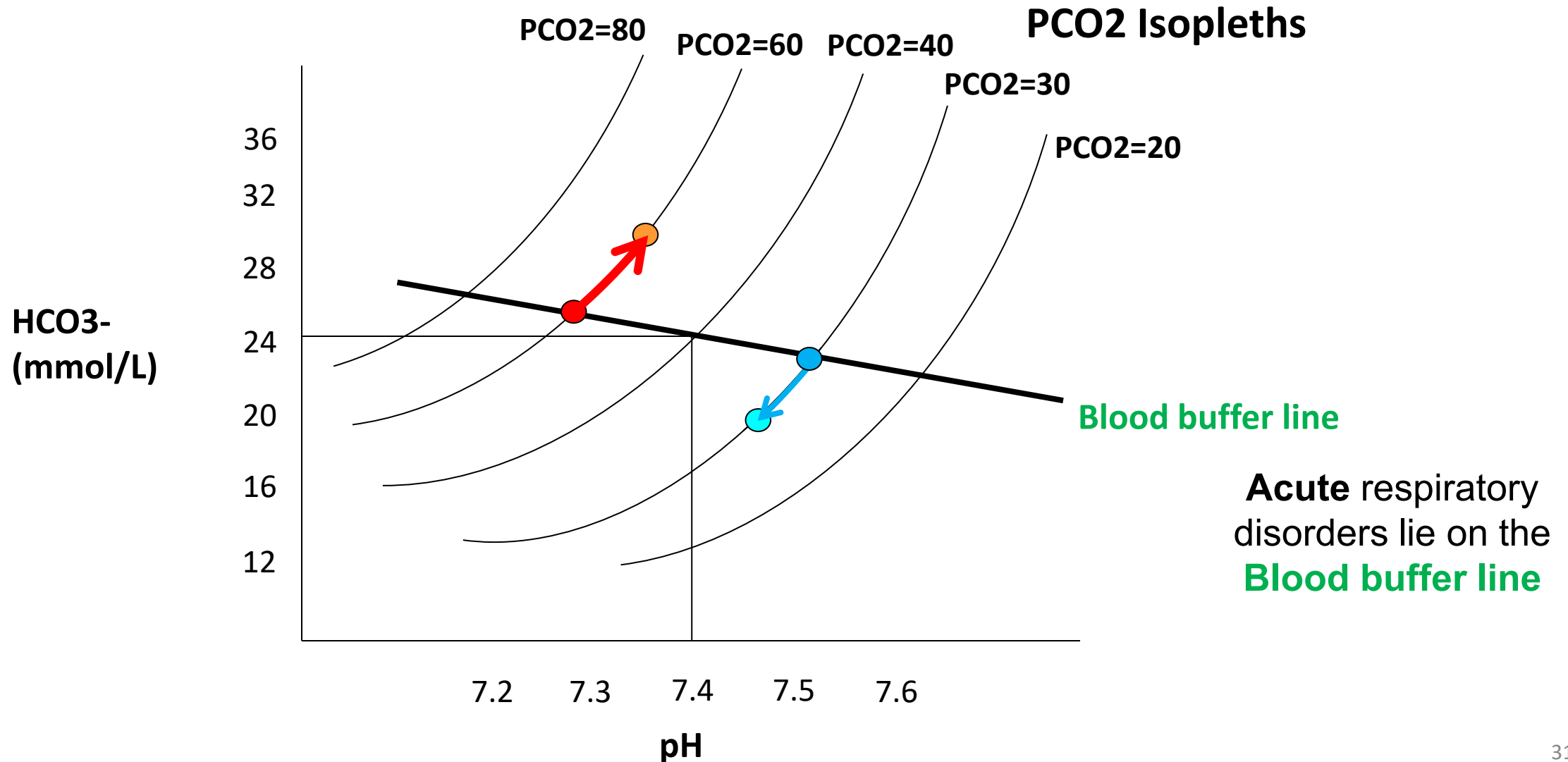


Chronic respiratory alkalosis

PCO₂ Isopleths

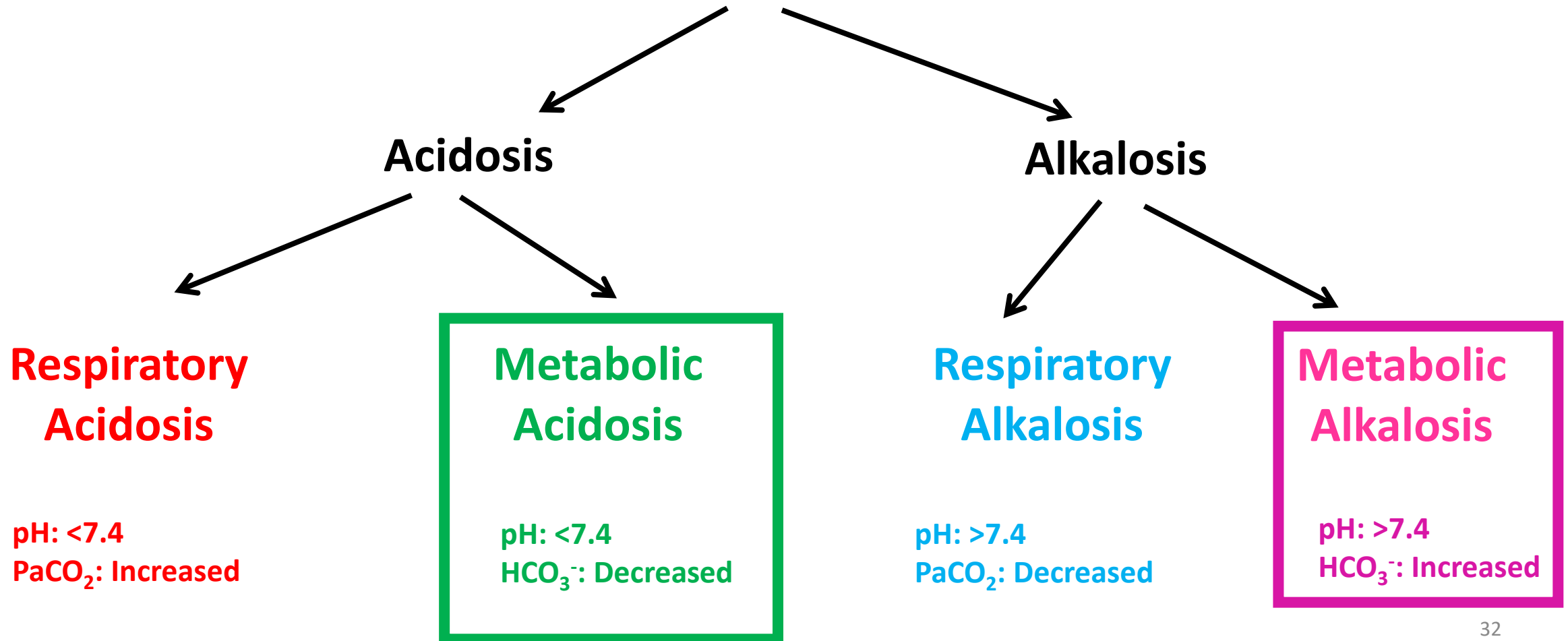


Respiratory acid-base disorders

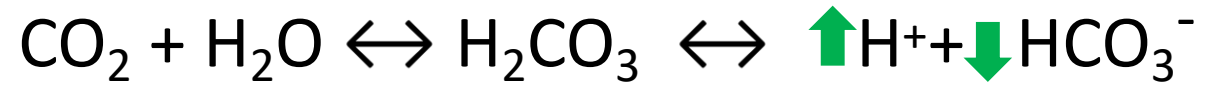


Classification of Acid-base disorders

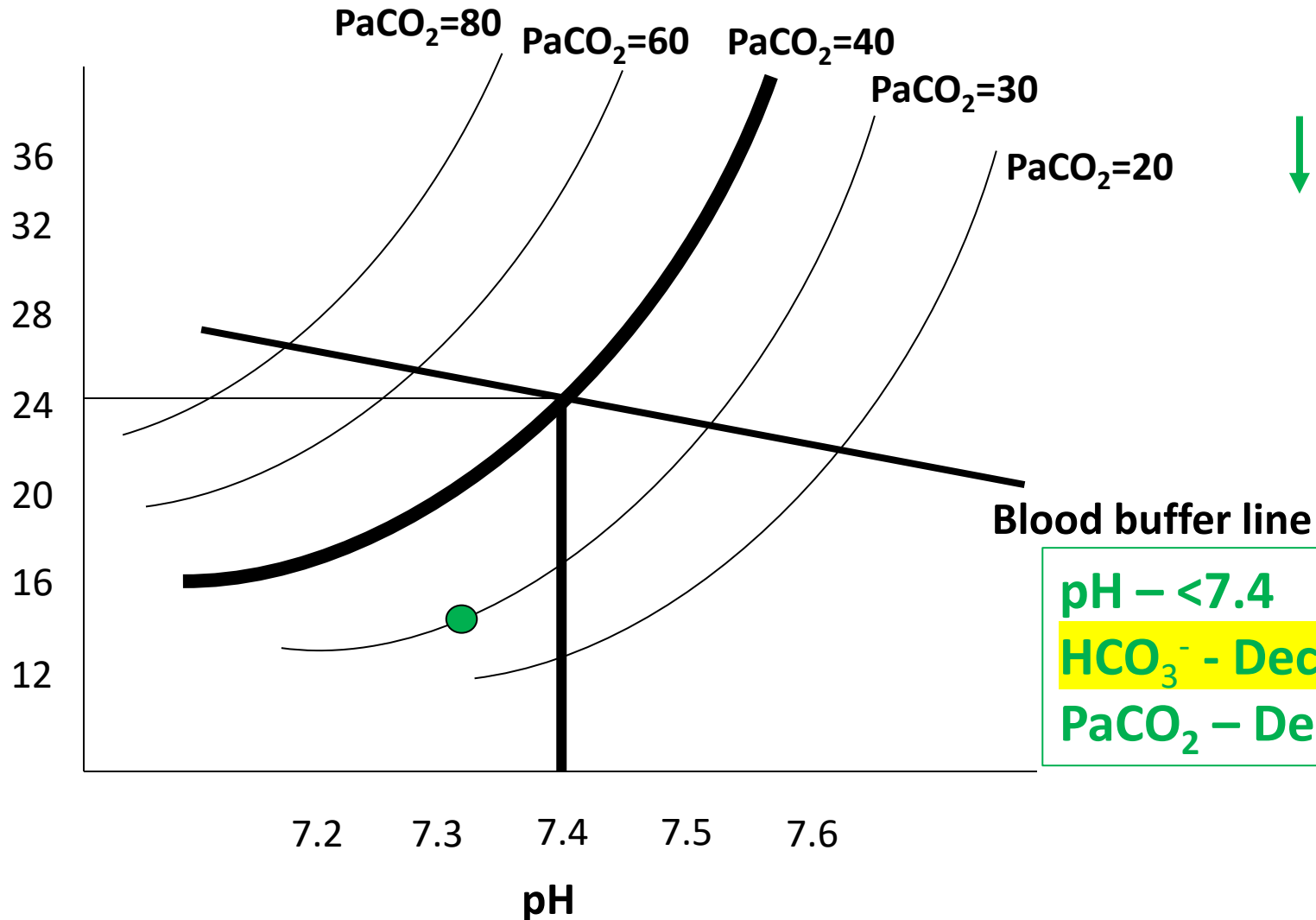
Based on pH



Metabolic acidosis ($\downarrow\downarrow\text{HCO}_3^-$)



HCO_3^-
(mmol/L)



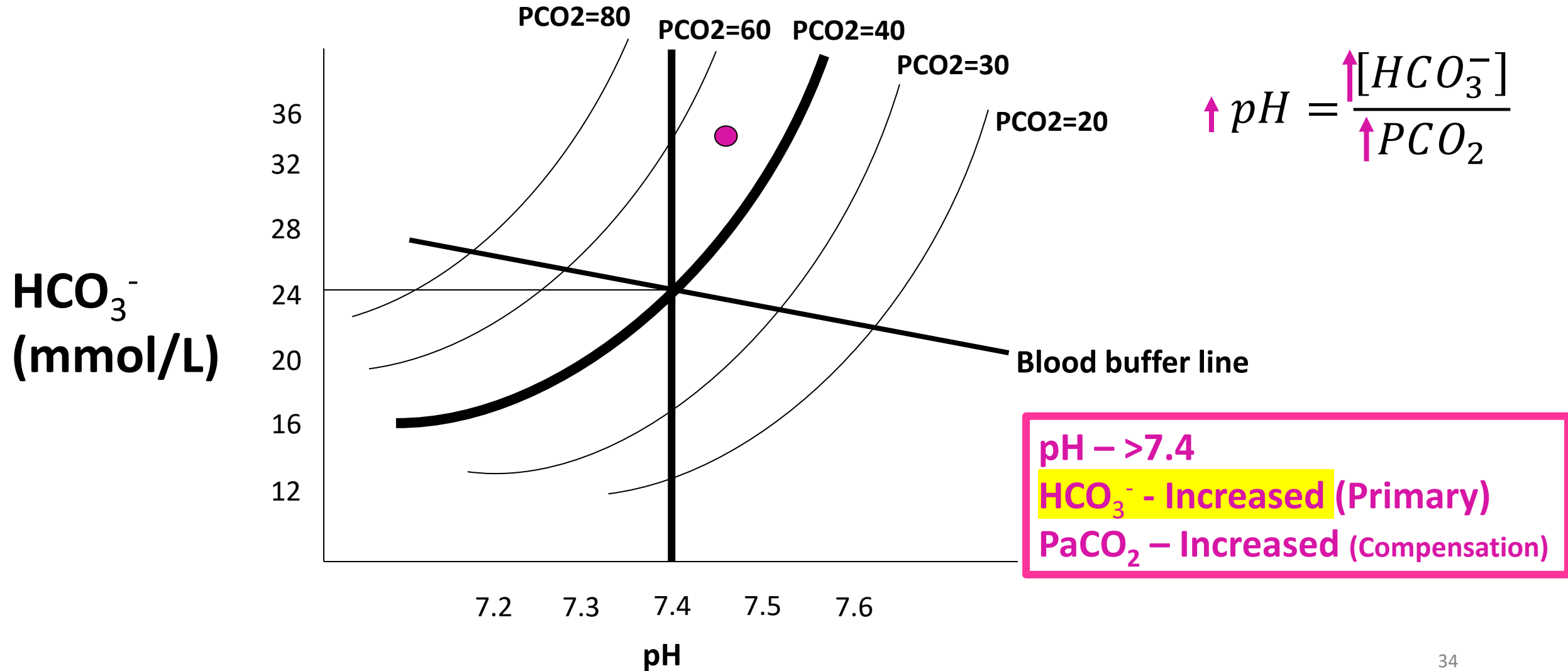
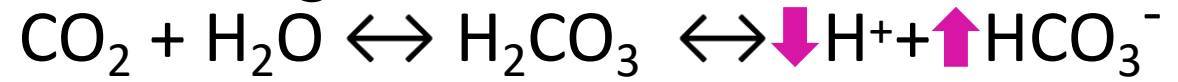
$$\downarrow pH = \frac{\downarrow [\text{HCO}_3^-]}{\downarrow \text{PCO}_2}$$

pH – <7.4

HCO_3^- - Decreased (Primary)

PaCO₂ – Decreased (Compensation)

Metabolic alkalosis ($\uparrow \uparrow \text{HCO}_3^-$)



Davenport Diagram

